

Facility Trust Desk

Receipts, not invented confidence.

Hack-Nation Challenge 04 · Data Legend · Facility Trust Desk track

Stephen Pasco (solo) · Live on Databricks Free Edition

`trustdesk-spike-7474651147859378.aws.databricksapps.com` · `github.com/timezyme/hacknation-6-global-hackathon`

The problem — a postal code often decides a lifespan

10,088 records

Indian medical facilities listing what they can do: ICU, maternity, trauma. Nobody has checked whether the claims are true.

No ground truth

Any confident score would be invented. The brief asks apps to double-check their own work instead.

Empty ≠ absent

Only 25% of records list capacity. Missing paperwork must never read as a missing hospital.

Trust Desk answers one question: does a facility's record support what it claims — with the evidence shown, and uncertainty stated honestly.

The planner's workflow — ranked by record support

Pick a capability and region. 7 ranked candidates, 60 shown honestly outside the ranking. Never facility quality.

■ FACILITY TRUST DESK Does a facility's record support what it claims?

Live bounded slice · free checks only

Record-based assessment. This tool reads what each facility's record says. It has not independently confirmed current clinical capability, staffing, or whether equipment is in working order.

STEP 1

What capability do you need?

ICU maternity emergency
oncology trauma NICU

STEP 2

Where?

Andhra Pradesh

READING THE EVIDENCE MARKS

- ☒ **Backs the claim.** This field says something that supports it.
- ☐ **Unresolved.** The free checks could not settle this field.
- ☒ **No data at all.** Field is empty. Absence of proof, not proof of absence.
- ☒ **Contradicts.** This field says the opposite of the claim.
- ☐ **Could not be read.** The record itself is damaged. This is our failure, not the facility's.

Records claiming ICU · Andhra Pradesh

7 ranked candidates · 60 shown outside the ranking Ordered by strength and consistency of the record — not by facility quality.

	Facility	Description	Capability	Equipment	Procedure	Assessment
01	Q1 Hospitals Andhra Pradesh - record record:41fe10a53c11459e0533e724ebaeb0ee18712d608c463e6da7d6dc3c1173858f:ce5ad0ed341dc15fa8f3e33440858fa10a0e1f0a3da42443e071425	☒	☒	☒	☐	Strong record support
02	Aarka Hospitals Andhra Pradesh - record record:39aa0bb3e26effdd740b6e8d27559788b8a7b599769fda57d251a23f3a648418:c18279e019e069fb3c3e506d8bbdd2842766d78b1bcc6eff9f98a	☒	☒	☒	☐	Strong record support
03	Aragonda Apollo Hospital Andhra Pradesh - record record:9bec7ce8c5417408897ed3370735c397b8babd2ad3e4d0f151fc665aaab79937:ecd3a265b9c5ab0ad58fa6c92af429a7789dc35ed28e46b724c39	☐	☒	☒	☒	Strong record support
04	Bharathi Hospital Andhra Pradesh - record record:e4d9b1325dff722bfbb4d18ee4a9f62ea67413e9d83075a7876797e6d627888d:fd81d5ad2b60e2cd2420cc4372a804dc07e2ecaa93253b443c07c	☐	☒	☒	☒	Strong record support
05	Sudarshan Hospitals Andhra Pradesh - record record:d4437c9133f0d03c1295c9f825555f95541ca5ab451b3ce89c29a7e1933232c7:3603997c1f8d66f0f004d34ca0dbca5cb6a5421f1afa4c4a996cc2	☐	☒	☒	☒	Strong record support
06	MJ Naidu Super Speciality Hospital Andhra Pradesh - record record:c1b53e5030d13bd4641277aefdd8f25de10fe9d7901b6c1db9de8f1b4efdd6070:24a81d659168c036d4ca470e5add1106bfac70ac5155a9ca7ec7cca	☒	☒	☒	☐	Strong record support

The receipt — evidence, grades, deciding check

Exact sentences, four field grades, check identity and version, attempt trail, and a second opinion per decision.

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Records claiming ICU - Andhra Pradesh

7 ranked candidates - 60 shown outside the ranking Ordered by strength and consistency of the record — not by facility quality.

Facility	Description	Capability	Equipment	Procedure	Assessment
01 Q1 Hospitals					
Andhra Pradesh - record					Strong record support
record:41fe10a53c11459e0533e724ebaeb0ee187f2d608c463e6da7d6dc3c1173858f:ce5ad0ed341dc15fa8f3e33440858fa10a0e1f0a3da42443e071425					

Claim assessed: Facility provides ICU care

Record

record:41fe10a53c11459e0533e724ebaeb0ee187f2d608c463e6da7d6dc3c1173858f:ce5ad0ed341dc15fa8f3e33440858fa10a0e1f0a3da42443e071425
· live India 10k slice · exact items below

Why this preview: 3 fields contain explicit free-check support, across 9 evidence items.

■ description [0]

Q1 Hospitals is a brain child promoted by reputed Joint Replacement Surgeon Dr. Ramana Murthy Tamminana, with an intend to setup a first ever surgical specialty hospital in the coastal Andhra Pradesh providing Quality and finest Health care services that is accessible and affordable. 7 State of Art Laminar Flow Modular Operation Theaters, Ultra-Modern Surgical Intensive Care Units defines us remarkable. Q1 stands for advancing the "Quality and Quick" healthcare services by facilitating total patient satisfaction through effective internal process controls. Our modern infrastructural advancements foster the service steadiness and effective performance to meet the growing future needs at reliable and affordable cost.

Honest states — a data desert is not a medical desert

'Not enough evidence' is shown, labelled, and never given a misleading low rank. Says-nothing and blank stay distinct.

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► How the assessment is decided

Bharathi Hospital

04 Andhra Pradesh - record

record:e4d9b1325dff722bfb4d18ee4a9f62ea67413e9d83075a7876797e6d627888d:fd81d5ad2b60e2cd2420cc4372a804dc07e2ecaa93253b443c07c

Strong record support

Sudarshan Hospitals

05 Andhra Pradesh - record

record:d4437c9133f0d03c1295c9f825555f95541ca5ab451b3ce89c29a7e1933232c7:3603997c1f8d66f0f004d34ca0dbca5cb6a5421f1afa4c4a996cc2d

Strong record support

MJ Naidu Super Speciality Hospital

06 Andhra Pradesh - record

record:c1b53e5030d13bd4641277aefdd8f25de10fe9d7901b6c1db9de8f1b4efd6070:24a81d659168c036d4ca470e5add1106bfac70ac5155a9ca7ec7cca

Strong record support

Swaasthya Hospital

07 Andhra Pradesh - record

record:ca416ef66348695e9601d79864d74081ad53552571c7f6c021c87e67cc60b22c:2fb8d9a39b38f4f2e55e20e88e7e77084fe717657156d46333c59f

Strong record support

Not enough evidence - 60

Too little data to judge. A data desert is not a medical desert. Outside the ranking on purpose — never a misleading low rank.

Aditya Hospitals

— Andhra Pradesh - record

record:4a54fb6c1eaa3c9a87761b1d6369a3a59f77bf19ea094580e2a9b67d5975cec:c2549059aa0789ad2fb80e622b87c6292729e1398df9402dce60a

Not enough evidence

Aditya Nursing Home

— Andhra Pradesh - record

record:1ff80c638b77647cd22af35b8257f712eddd3a063a45417c53700509549704e0:4e6a74e2a3bcf57c30945e54fb36009cf7a56caf3bcbf30e47a91c

Not enough evidence

Ahalya Nursing Home, Guntur

— Andhra Pradesh - record

record:627e4b4c821e530b9ba59af8eb29a69c9ca5cb049e7a0eed3530ccbec41ec129:d5510040cc7e39fcc3e90b104bfe887f943122421cb5b364deae4

Not enough evidence

Amara Hospital

— Andhra Pradesh - record

record:9fef652ef8629c1f8fde08cf9751e5f679216bf7617e4187a5b1590314f83a62:30134671c651627d21a20a658a5148500e9d05fce6e67b20476d4fa5

Not enough evidence

Overrides — human knowledge persists

Confirm or override with a note. Snapshotted with the verdict it disagreed with; survives restarts, stored in Lakebase.

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► How the assessment is decided

Row-level source set

1. [Row source 1](#)
2. [Row source 2](#)
3. [Row source 3](#)
4. [Row source 4](#)
5. [Row source 5](#)
6. [Row source 6](#)
7. [Row source 7](#)
8. [Row source 8](#)
9. [Row source 9](#)
10. [Row source 10](#)
11. [Row source 11](#)

Still unknown: 86 evidence item(s) unresolved by the configured checks.

OVERRIDDEN BY YOU · PERSISTED

Our field team visited in June: ICU is real but only two beds.

Edit note

Aarka Hospitals

02 Andhra Pradesh · record

record:39aa0bb3e26effdd740b6e8d27559788b8a7b599769fda57d251a23f3a648418:c18279e019e069fb3c3e506d8bbdd2842766d78b1bcc6eff9f98a3

Aragonda Apollo Hospital

03 Andhra Pradesh · record

record:9bec7ce8c5417408897ed3370735c397b8babd2ad3e4d0f151fc665aaab79937:ecd3a265b9c5ab0ad58fa6c92af429a7789dc35ed28e46b724c39

Bharathi Hospital

04 Andhra Pradesh · record

record:e4d9b1325dff722bfbb4d18ee4a9f62ea67413e9d83075a7876797e6d627888d:fd81d5ad2b60e2cd2420cc4372a804dc07e2ecaa93253b443c07c

Measurements — how much to trust each check

Blind-labelled pilot: per-check coverage and precision with 95% confidence intervals. Reviewer feedback is never called accuracy.

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About these ranks

Unresolved evidence stays visible. These ranks describe record support, not facility quality or current clinical capability.

Measurements — how much to trust each check

Pilot numbers are evidence-label agreement on a blind sample, per check, with confidence intervals. Reviewer feedback is never counted as accuracy.

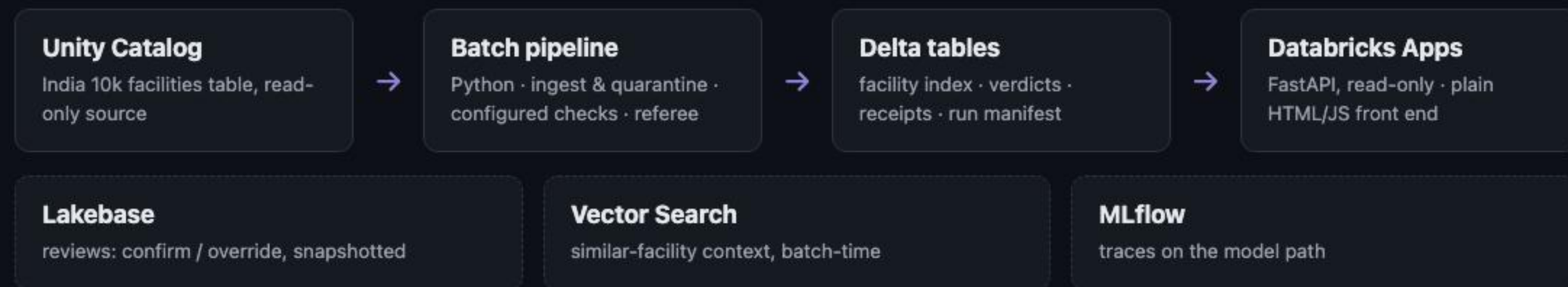
development split

Capability	Check	Coverage	95% CI	Precision	95% CI	Errors
ICU	presence	0%	0–39%	—	—	0
ICU	vocabulary	0%	0–39%	—	—	0
NICU	presence	0%	0–39%	—	—	0
NICU	vocabulary	17%	3–56%	100%	21–100%	0
emergency	presence	0%	0–39%	—	—	0
emergency	vocabulary	17%	3–56%	100%	21–100%	0
maternity	presence	0%	0–39%	—	—	0
maternity	vocabulary	17%	3–56%	100%	21–100%	0
oncology	presence	0%	0–39%	—	—	0
oncology	vocabulary	0%	0–39%	—	—	0
trauma	presence	0%	0–39%	—	—	0
trauma	vocabulary	0%	0–39%	—	—	0

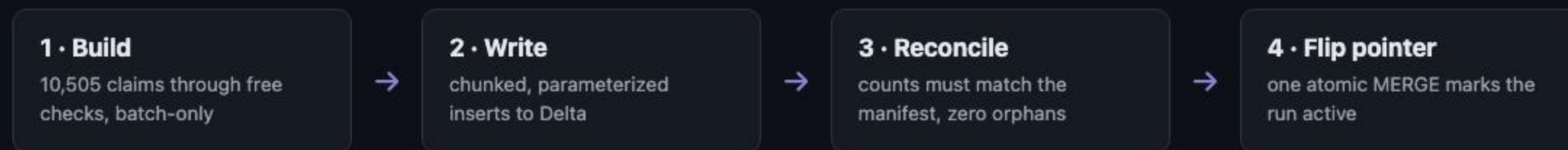
holdout split

Capability	Check	Coverage	95% CI	Precision	95% CI	Errors
ICU	presence	0%	0–49%	—	—	0
ICU	vocabulary	0%	0–49%	—	—	0
NICU	presence	0%	0–49%	—	—	0
NICU	vocabulary	0%	0–49%	—	—	0
emergency	presence	0%	0–49%	—	—	0
emergency	vocabulary	0%	0–49%	—	—	0

Stack — all Databricks Free Edition



Architecture — nothing runs while a planner waits



A partial batch can never go live. The previous completed run stays queryable for rollback. One 2X-Small warehouse is enough, because the app only ever reads.

Implementation — checks are plug-ins

```
# config/checks.toml
checks = [
  "trustdesk.check_presence:PresenceCheck",
  "trustdesk.check_vocabulary:VocabularyCheck",
]
[referee]
enabled = true
mode = "rules"
```

```
# the whole check interface
class Check(Protocol):
    check_id: str
    implementation_version: str
    cost_tier: CostTier
    def evaluate(evidence)
        -> tuple[CheckFinding, ...]
```

Ordered by cost, cheapest first. A check may **decide, abstain, or fail** — never guess. A regression test proves a new check needs no pipeline changes.

Trust mechanics — the app double-checks itself

Referee

every decision re-examined by an independent method: agree · disagree · not double-checked. Shared wording is masked, so a term can never confirm itself.

Vector Search

most similar facility records attached to ranked receipts — labelled comparison context, not verification

Verdict rule

five field grades reduce by a fixed, visible rule. No model ever writes a verdict.

Honest states are first-class: says-nothing ≠ blank, failures are never "no evidence", and unranked states are shown, never scored low.

Shipped and verified

```
$ uv run pytest --cov=trustdesk --cov=app
163 passed · coverage 81% (gate: 80)

$ ruff check src app tests scripts && mypy src app scripts
clean · clean

$ python scripts/smoke_demo.py           # against the deployed app
health 200 · 6 capabilities · receipt: trail + referee + similar
review round trip: pass · cold restart: reviews intact
```

github.com/timezyme/hacknation-6-global-hackathon · live on Databricks Free Edition

The brief's research questions — answered in the shipped app

Trust without ground truth

Per-check coverage and precision with 95% confidence intervals, from a blind-labelled pilot with a sealed holdout — shown in the app.

Claims vs. evidence

Four field grades separate a corroborated ICU claim from a bare listing. Corroborated claims outrank bare ones.

The data desert problem

"Says nothing" and "blank" are distinct grades, shown as honest states — never merged, never ranked low.

Plus the validator stretch goal: an independent referee re-examines every decision, and disagreement is displayed, never hidden.